Sound-Based Emergency Event Detection

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Motivation

Main Idea

- A smart sensing system that recognize the Emergency Sound before user visualize them.

Possible Events

- Fire Alarm
- Siren
- Glass Breaking
- Screaming
- Loud Impact sounds
- Smoke Alarm

Why it Matters

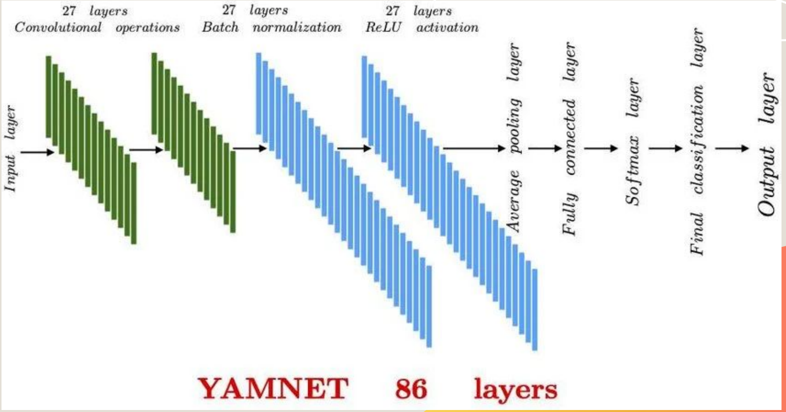
- It helps in improving the smart home security system
- Support the elderly people or people living alone
- Useful for people with hearing difficulties

ML Component

Task: Multi-class emergency sound classification

Pretrained Model: YAMNet

- The YAMNet is a pretrained audio event classification model trained on AudioSet



Possible Classes

- Fire Alarm / Smoke Alarm
- Siren
- Glass Breaking
- Normal Background Noise
- Scream

Model Design

Audio Waveform -> Pretrained YAMNet -> Audio Embeddings -> Classification Layers -> Labels

Sensing Component

Sensor Used: Microphone from laptop or phone

Input Data

Continuous audio stream, processed in 1 second windows

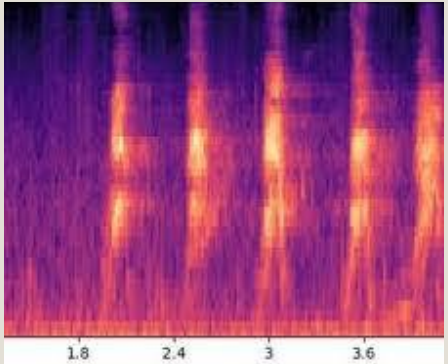
Captured at 16kHz sample rate

Preprocessing

Split audio into fixed-size overlapping windows

Convert each window into a mel spectrogram

Normalize input for consistent model performance



Microphone → Audio Windows → Mel Spectrogram → Model → Notification

Evaluation

Category	Metric	Success Criteria
Classification	F1-Score Mean of precision and recall	**> 0.85** emergency classes
Timing	Detection Latency Processing time from sound to alert	**< 1.5 seconds** alert trigger
Robustness	False Alarm Rate % of non-emergency sounds flagged	**< 2%** normal background
Sensing	Sensitivity Effective range for clear recognition	**Up to 10 meters** detection

Live Demo Plan

Environment Setup

- Laptop running dashboard
- Sound sources, such as emergency smoke alarm and background noise

The Workflow

Mic → Mel-Spectrogram → Model → UI Alert

Components

- Real-time visualization of audio waveform
- Visual/Sound notification when alarm triggers

Any questions?